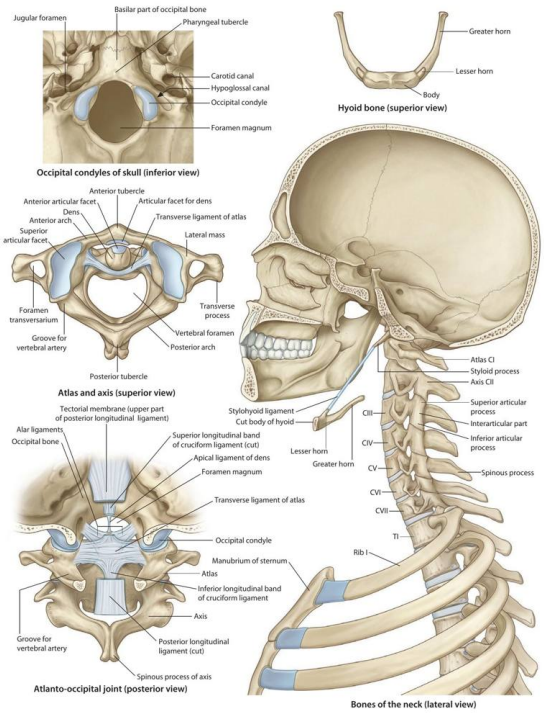
Neck bones

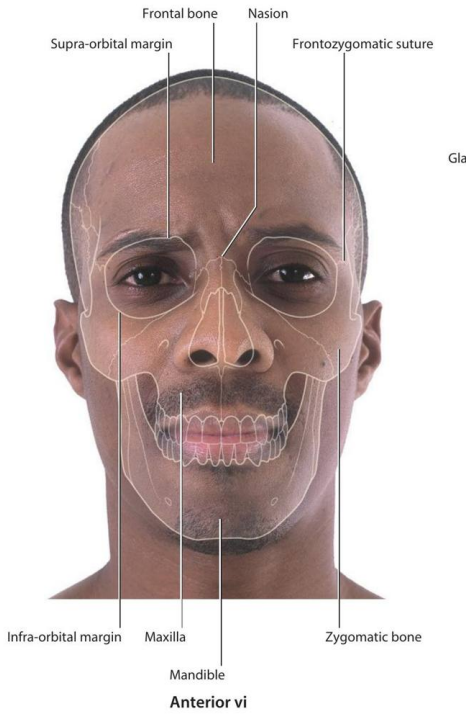
Hyoid and cervical vertebrae: how to identify each one, what attaches to it, and the vertebral levels you use for ENT surface anatomy.

Vertebral levels and surface landmarks

- **Hyoid:** C3. **Upper border of the thyroid cartilage:** C4, near the common carotid bifurcation.
- **Cricoid cartilage:** C6. The larynx becomes the trachea and the pharynx becomes the oesophagus here.
- **Transverse process of the atlas:** deep to the point between the mastoid tip and the angle of the mandible.
- **C7:** the first clearly palpable spine at the base of the neck (vertebra prominens). Levels vary with age and neck position.



Bones of the neck: hyoid, atlas, axis and cervical spine. Gray's Atlas of Anatomy, 2nd ed., p. 537

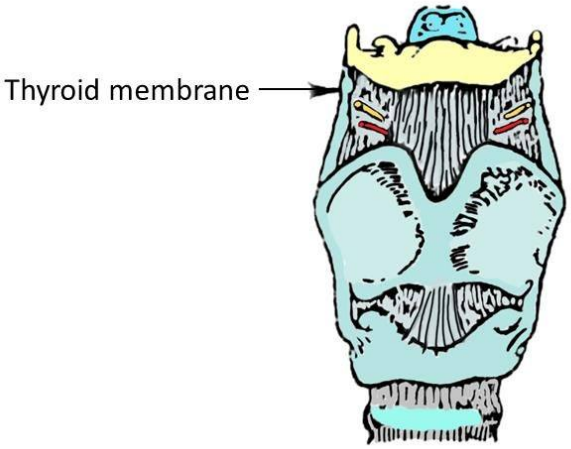


Surface anatomy with bones: anterior view. Gray's Atlas of Anatomy, 2nd ed., p. 476

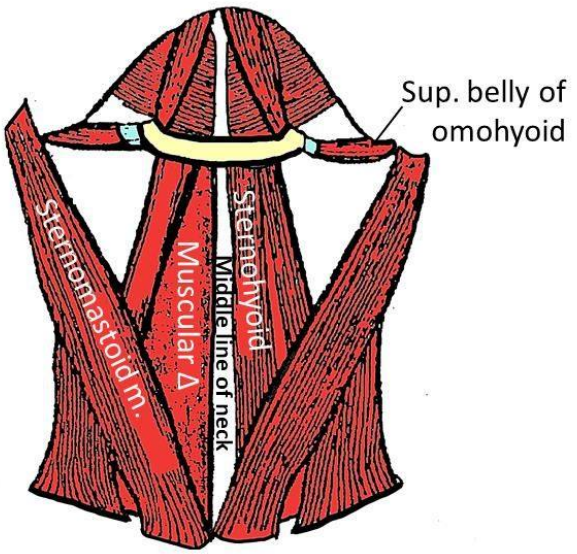
Hyoid bone

- **Parts:** body, two greater horns and two lesser horns. It forms **no bony joint** with another bone.
- **Embryology:** lesser horn and upper body from the **second arch**, greater horn and lower body from the **third arch**.
- **To orient it:** the body's anterior surface is convex and faces forward and up; its posterior surface is concave. The greater horns point backward.
- **Suspended by:** the stylohyoid ligament (to the lesser horn), and suprahyoid and infrahyoid muscles.

Clinical pearl: A thyroglossal cyst operation (Sistrunk) removes the central part of the hyoid body, because the tract is closely related to it.



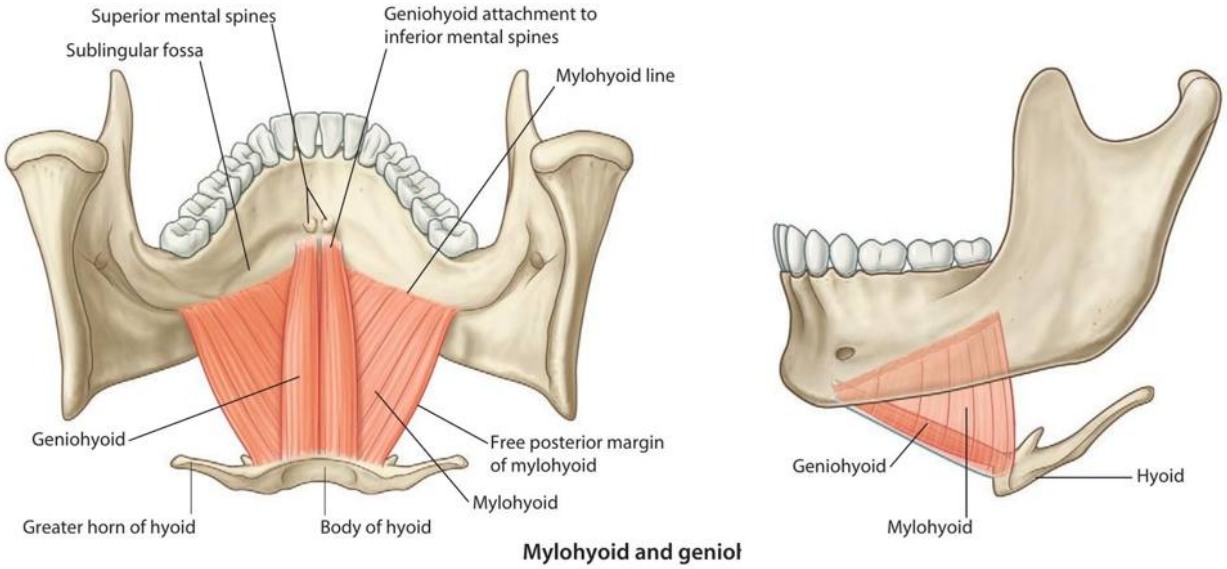
Thyrohyoid membrane. Dr. Sameh Doss handout, p. 172



Infrahyoid strap muscles. Dr. Sameh Doss handout, p. 122

Muscles and membranes on the hyoid

- **Front of the body:** geniohyoid, mylohyoid; stylohyoid at the junction of body and greater horn.
- **Lower border of the body:** sternohyoid and omohyoid. **Lower body and greater horn:** thyrohyoid.
- **Greater horn:** hyoglossus and middle constrictor. **Lesser horn:** middle constrictor and stylohyoid ligament.
- **Digastric:** its intermediate tendon is held to the body and greater horn by a fibrous sling.
- **Thyrohyoid membrane:** attaches to the upper border of the **back of the body** and greater horn, leaving a bursa behind the body.

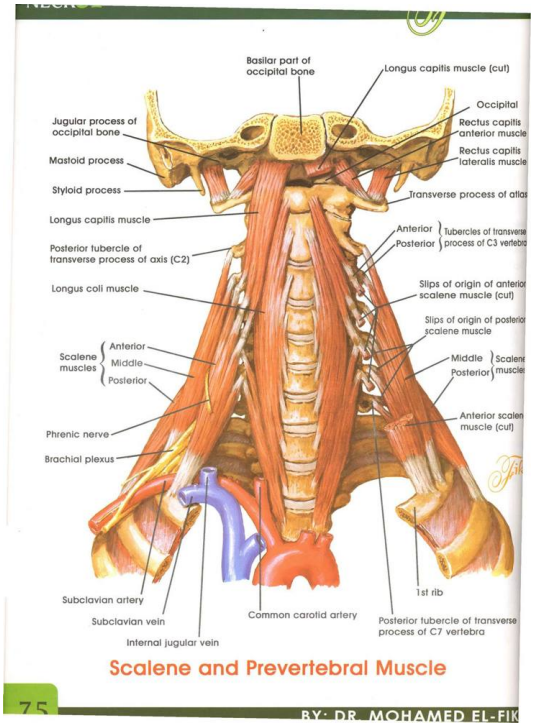


Mylohyoid and geniohyoid. Gray's Atlas of Anatomy, 2nd ed., p. 578

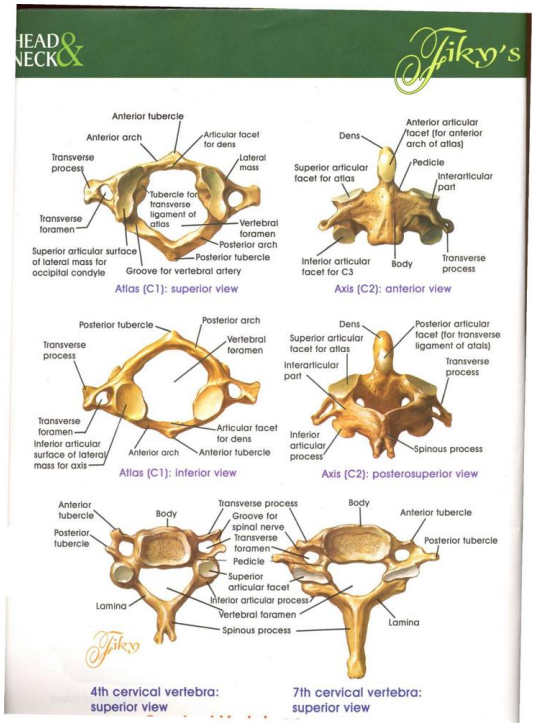
Typical cervical vertebra (C3–C6)

- **Body:** small and oval, with raised lips (uncinate processes) at the sides of its upper surface.
- **Foramen transversarium:** in every transverse process. This identifies a vertebra as cervical.
- **Transverse process:** anterior and posterior tubercles. **Spinous process:** short and usually bifid. **Vertebral foramen:** large and triangular.
- **Attachments:** anterior scalene from the anterior tubercles of C3–C6; middle scalene from the posterior tubercles; levator scapulae from C1–C4.

Clinical pearl: The anterior tubercle of C6 (carotid tubercle) is where the common carotid can be pressed against bone.



Scalene and prevertebral muscles. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 75



Cervical vertebrae: atlas, axis, C4 and C7. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 11

Atlas (C1)

- **No body and no spinous process.** It is a ring with an **anterior arch, a posterior arch and two lateral masses**.
- **Superior articular facets:** concave, for the occipital condyles (atlanto-occipital joint: nodding).
- **Groove on the posterior arch:** vertebral artery and C1 nerve. **Inner anterior arch:** facet for the dens.
- **Transverse process:** long and wide. Its tip is a surgical landmark near the accessory nerve and internal jugular vein.

Clinical pearl: The anterior arch of the atlas lies behind the nasopharynx at about the level of the hard palate.

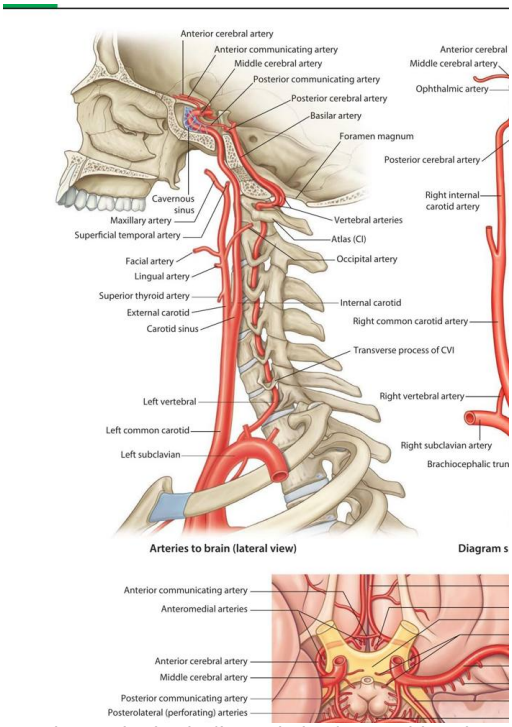
Axis (C2)

- **Dens (odontoid process):** projects up from the body and is held against the atlas by the **transverse ligament of the atlas**.
- **Ligaments of the dens:** apical to the foramen magnum; alar to the occipital condyles.
- **Spinous process:** large and bifid. **Atlanto-axial joints:** rotation of the head (shaking).
- **Foramen transversarium** is angled to pass the vertebral artery laterally towards C1.

Clinical pearl: Grisel's syndrome is atlanto-axial subluxation following pharyngeal infection or adenotonsillectomy. Think of it in a child with torticollis after surgery.

C7 and the vertebral artery

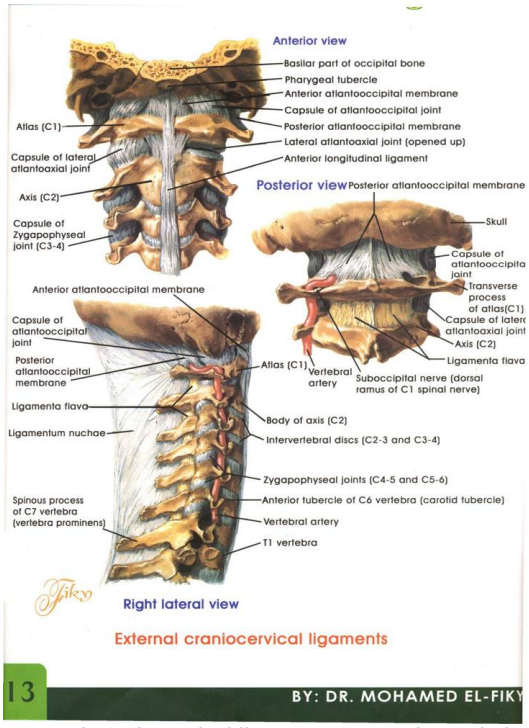
- **C7 (vertebra prominens):** a long spine that is **not bifid**, with a tubercle at its tip for the ligamentum nuchae.
- **Vertebral artery:** normally enters at the foramen transversarium of **C6**, not C7, and runs up through C1.
- **Foramen of C7:** small; usually transmits only accessory vertebral veins.
- **Cervical rib:** can arise from C7 and may compress the lower brachial plexus or subclavian artery.



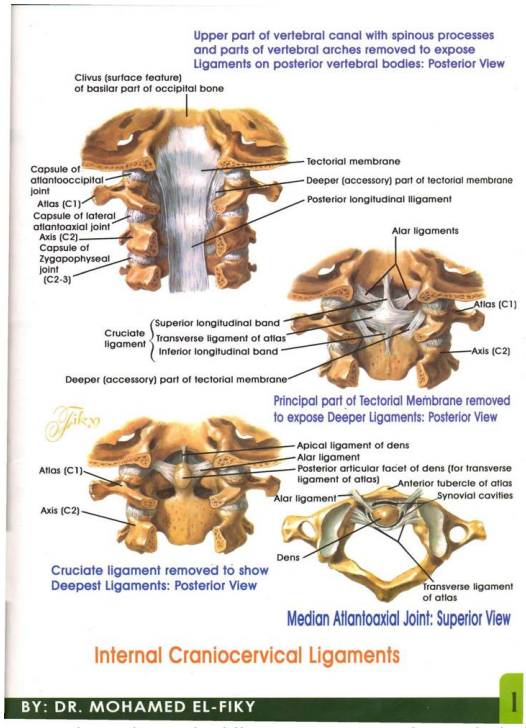
Arteries to the brain (lateral view): carotid and vertebral systems. Gray's Atlas of Anatomy, 2nd ed., p. 500

Craniocervical ligaments

- **Tectorial membrane:** the upward continuation of the posterior longitudinal ligament to the clivus.
- **Cruciate ligament:** the transverse ligament of the atlas with superior and inferior longitudinal bands.
- **Alar ligaments:** dens to the occipital condyles. **Apical ligament:** dens to the foramen magnum.
- **Anterior atlanto-occipital membrane:** continuous below with the anterior longitudinal ligament. **Posterior atlanto-occipital membrane:** the vertebral artery and suboccipital nerve pass beneath its edge.
- **Behind the spine:** ligamenta flava and ligamentum nuchae.



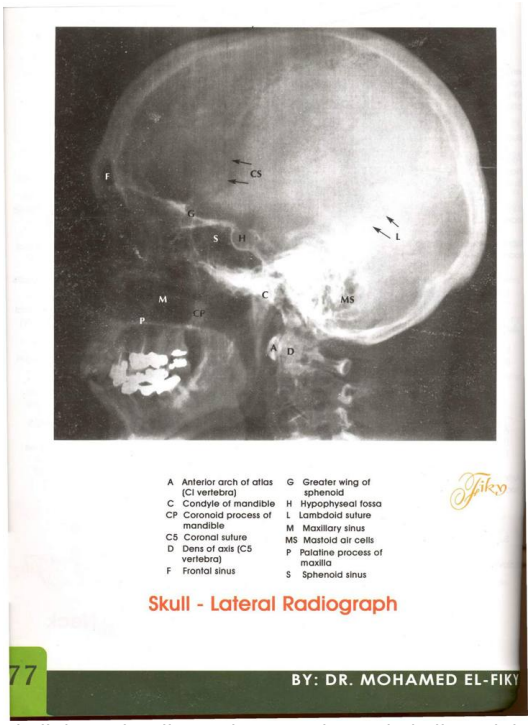
External craniocervical ligaments. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 13



Internal craniocervical ligaments. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 14

On imaging

- **Lateral skull radiograph:** look for the anterior arch of the atlas in front of the dens.
- The gap between the two is widened in atlanto-axial subluxation.



Skull, lateral radiograph. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 77

In theatre

- **Greater horn of the hyoid:** a neck dissection landmark. The hypoglossal nerve runs just above it and the lingual artery arises near its tip.
- **Transverse process of the atlas:** helps locate the accessory nerve and internal jugular vein high in the neck.
- **Sistrunk operation:** removes the central part of the hyoid body with the thyroglossal tract.

Attachments on the hyoid

Site	Attachment	Nerve
Front of body	Geniohyoid	C1 via XII
Body	Mylohyoid	Nerve to mylohyoid (V3)
Body and greater horn junction	Stylohyoid	VII
Lower border of body	Sternohyoid	Ansa cervicalis
Lower border of body, lateral	Omohyoid	Ansa cervicalis
Greater horn and lower body	Thyrohyoid	C1 via XII
Greater horn	Hyoglossus	XII
Greater and lesser horns	Middle constrictor	Pharyngeal plexus (X)
Lesser horn	Stylohyoid ligament	Not applicable

Telling cervical vertebrae apart

Vertebra	Key feature	Spine
Atlas (C1)	Ring with no body	No spine; posterior tubercle
Axis (C2)	Dens	Large and bifid
Typical (C3–C6)	Foramen transversarium, small oval body	Short and usually bifid
C7	Vertebra prominens	Long and not bifid

Personal study use.